

PUKAPUKA, Cook Islands

WMO: 918110

Lat: 10.915S

Lon: 165.839W

Elev: 15

StdP: 14.69

Time Zone: -10.00 (W10)

Period: 86-02

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	75.1	76.3	69.9	109.9	81.3	71.5	116.2	81.4	25.8	80.1	23.0	81.7	6.2	90	0.480

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	7.8	90.5	80.9	89.9	80.7	89.3	80.5	82.8	87.8	82.3	87.5	81.8	87.0	8.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.3	163.0	86.4	80.8	160.1	85.9	80.3	157.5	85.5	46.8	88.6	46.1	87.9	45.5	87.2	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.6	20.6	19.2	62.8	92.3	3.3	1.2	60.5	93.1	58.5	93.8	56.7	94.5	54.3	95.4		
(5)			60.0	83.8	4.8	1.4	56.6	84.8	53.8	85.7	51.1	86.4	47.7	87.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	83.3	83.7	83.3	83.4	84.0	84.0	83.3	82.4	82.6	83.0	83.2	83.3	83.1	(6)
(7)		DBStd	1.96	1.73	2.05	2.06	1.79	1.94	1.90	1.83	1.64	1.90	1.85	1.97	2.14	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	12141	1044	932	1036	1021	1053	999	1004	1010	989	1030	998	1025	(10)
(11)		CDD65	6666	579	512	571	571	588	549	539	545	539	565	548	560	(11)
(12)		CDH74	76020	6695	5836	6536	6716	6985	6341	5942	5972	6036	6410	6252	6299	(12)
(13)		CDH80	25633	2368	2003	2306	2530	2629	2150	1718	1711	1894	2127	2127	2070	(13)
(14)	Wind	WSAvg	9.3	7.6	8.6	8.1	7.1	9.1	11.4	11.5	11.7	11.2	9.2	8.1	8.6	(14)
(15)	Precipitation	PrecAvg	109.8	14.0	12.5	10.3	5.8	7.5	7.1	6.4	6.1	6.9	7.6	10.6	14.6	(15)
(16)		PrecMax	153.9	33.9	27.0	22.8	15.5	17.6	14.7	19.5	15.2	15.2	19.2	19.6	29.8	(16)
(17)		PrecMin	76.8	5.2	5.3	3.1	0.5	1.6	1.7	0.5	0.3	0.4	1.7	4.3	3.1	(17)
(18)		PrecStd	19.2	7.1	6.3	5.3	3.3	3.6	3.4	4.6	3.5	3.3	3.9	3.9	5.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.7	90.8	90.9	90.8	90.5	89.4	88.8	88.6	89.7	90.3	90.5	90.6	(19)
(20)			MCWB	80.6	80.9	81.3	81.8	81.4	80.6	79.1	79.4	80.0	79.9	80.7	81.2	(20)
(21)		2%	DB	89.7	89.7	89.9	89.9	89.5	88.3	87.6	87.5	88.6	89.2	89.3	89.4	(21)
(22)			MCWB	80.3	80.5	81.0	81.3	81.0	80.2	79.3	79.2	79.6	79.6	80.2	80.7	(22)
(23)		5%	DB	88.9	88.7	88.9	89.1	88.7	87.5	86.7	86.6	87.6	88.2	88.5	88.4	(23)
(24)			MCWB	80.1	80.3	80.7	81.0	80.8	79.9	79.1	79.1	79.3	79.5	80.0	80.4	(24)
(25)		10%	DB	87.9	87.6	87.9	88.0	87.7	86.6	85.8	85.7	86.5	87.2	87.4	87.3	(25)
(26)			MCWB	80.0	80.0	80.5	80.7	80.7	79.7	78.9	78.9	79.1	79.4	79.8	80.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	82.5	82.6	83.8	83.8	83.5	82.6	81.5	81.4	81.6	81.9	82.5	82.8	(27)
(28)			MADB	87.8	88.5	88.1	88.9	87.6	86.7	85.8	85.6	86.7	86.5	88.0	88.2	(28)
(29)		2%	WB	81.7	81.7	82.4	82.5	82.5	81.6	80.8	80.7	80.8	81.0	81.5	81.9	(29)
(30)			MADB	87.1	87.6	87.9	88.0	87.5	86.0	85.3	85.5	86.3	86.2	87.0	87.3	(30)
(31)		5%	WB	81.0	81.0	81.6	81.8	81.8	80.9	80.1	80.0	80.1	80.6	80.8	81.2	(31)
(32)			MADB	86.6	86.7	87.1	87.1	86.8	85.4	84.7	84.8	85.6	86.0	86.4	86.3	(32)
(33)		10%	WB	80.6	80.4	80.8	81.0	81.1	80.4	79.6	79.4	79.7	80.0	80.2	80.6	(33)
(34)			MADB	86.2	85.9	86.3	86.2	86.0	84.9	84.1	84.1	84.9	85.3	85.5	85.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.4	8.2	8.6	8.3	7.8	6.9	7.1	6.9	7.4	7.7	7.9	7.8	(35)
(36)			MCDBR	9.6	9.7	9.9	9.5	8.9	7.9	8.4	8.2	8.6	9.4	9.3	9.2	(36)
(37)			MCWBR	4.0	4.2	4.4	4.0	4.3	4.2	4.2	4.0	4.2	4.1	4.3	4.2	(37)
(38)		5% WB	MCDBR	8.5	8.3	8.9	8.6	8.0	6.9	7.0	7.3	7.5	7.9	8.4	8.3	(38)
(39)			MCWBR	4.1	4.0	4.8	4.3	4.4	4.4	4.2	4.1	4.1	4.2	4.4	4.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.387	0.392	0.379	0.382	0.386	0.392	0.389	0.395	0.398	0.382	0.382	0.383	(40)	
(41)		taud	2.629	2.608	2.656	2.653	2.636	2.597	2.598	2.576	2.568	2.625	2.639	2.642	(41)	
(42)		Ebn at Noon	303	299	298	288	277	270	273	280	288	299	302	304	(42)	
(43)		Edh at Noon	32	33	30	29	28	29	29	31	33	32	31	31	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1769	1794	1824	1767	1571	1493	1588	1732	1938	1959	1896	1874	(44)	
(45)		RadStd	176	248	143	140	94	79	103	107	120	137	149	158	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page